

Power BI Case Study – Superstore Sales Analysis (Beginner)

Business Scenario

A retail company named Superstore sells Office Supplies, Furniture, and Technology products across different regions. Management wants an interactive dashboard to analyze sales, profit, products, customers, and regional performance.

Objective

Create an interactive Power BI dashboard to analyze sales performance and answer business questions.

Dataset Information

Rows: 9,994

Columns: 21

Important Columns:

- Order Date
- Ship Date
- Ship Mode
- Customer Name
- Segment
- Country
- City
- State
- Region
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

Part 1 – Data Preparation (Power Query)

- Check data types
- Rename columns if required
- Remove duplicate rows
- Check for null values
- Convert Order Date and Ship Date to Date format
- Close & Apply

Part 2 – Data Modeling

- Create a Calendar Table
- Create relationship: Calendar[Date] → Orders[Order Date]
- Mark Calendar Table as Date Table

Part 3 – DAX Measures

- Total Sales
- Total Profit
- Total Orders (Distinct Count)
- Total Quantity Sold
- Average Sales
- Profit Margin %

Part 4 – Dashboard Requirements

KPI Cards:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity

Visuals:

1. Sales by Category (Bar Chart)
2. Profit by Sub-Category (Bar Chart)
3. Monthly Sales Trend (Line Chart)
4. Sales by Region (Donut Chart)
5. Sales by Segment (Pie Chart)
6. Top 10 Products by Sales (Bar Chart)
7. Profit by State (Map)
8. Order Details (Table)

Slicers:

- Region
- Category
- Segment
- Year

Business Questions

1. What is the total sales?
2. What is the total profit?
3. Which category generates the highest sales?
4. Which sub-category earns the highest profit?
5. Which region contributes the most sales?
6. Which customer segment generates the highest revenue?

7. Which state has the highest sales?
8. Which products generate the most revenue?
9. Which products are making losses?
10. How do sales change month by month?

Learning Outcomes

Students will learn:

- Power Query
- Data Modeling
- Calendar Table
- DAX Measures
- KPI Cards
- Charts & Maps
- Slicers
- Dashboard Design